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TATA

TATA INSIGHTS AND QUANTS

Impact Quantified

Data Visualisation

Empowering Business with Effective Insights

Forage Job Simulation — Comprehensive Report

Forage

Presented by

PGDM
Business Analytics
Student

Under the guidance of Prof. Ajit Kumar

Executive Summary

This report presents the findings from the TATA Insights and Quants (iQ) Data Visualisation job simulation on Forage. The simulation involved four key tasks: framing business scenarios, selecting appropriate visualisations, creating effective data visuals in Power BI, and communicating insights to senior leadership.

The analysis covered an online retail store dataset, focusing on revenue drivers, seasonal trends, customer behaviour, and geographic demand to support the CEO and CMO in strategic expansion planning.

Task 1: Framing the Business Scenario

Background

An online retail store engaged a consultant to analyse data and provide insights valuable to the CEO and CMO. The business performed well and management needed to identify major revenue-contributing factors to plan strategically for the next year.

CEO Questions

#	Question	Business Rationale
1	Which region generates the highest and lowest revenue?	Understand geographic revenue distribution for expansion decisions
2	What is the monthly revenue trend — where are the biggest spikes/dips?	Identify seasonal patterns and internal change impacts
3	Which months generate the most revenue? Is there seasonality?	Maximise peak-season strategy and forecast accurately
4	Who are the top customers and how dependent is the business on them?	Assess revenue concentration risk and diversification needs

CMO Questions

#	Question	Business Rationale
1	What % of customers are repeat buyers? What do they reorder?	Target repeat customers with personalised offers
2	How long do repeat customers take to place their next order?	Optimise re-engagement timing and incentive campaigns
3	What revenue comes from customers who have ordered more than once?	Quantify the value of loyalty and justify retention spend
4	Who repeats most frequently and what is their revenue contribution?	Distinguish high-frequency/low-value vs low-frequency/high-value segments

Task 2: Choosing the Right Visuals

Selecting the correct visualisation type is critical - incorrect representations can lead to wrong decisions by senior management.

Visual Selection Answers

Question	Scenario	Best Visual	Why
Q1	CEO: Time series of revenue for the full year with seasonal trends	Line Chart	Best for showing trends and fluctuations over continuous time
Q2	CMO: Top 10 countries by revenue with product breakdown	Stacked / Grouped Bar Chart	Shows both country totals and product sub-components in a single view
Q3	CEO: Average revenue per country with distribution stats (min, Q1, median, Q3, max)	Box Plot (Box & Whisker)	Designed specifically to display the five-number summary
Q4	CMO: Top 10 customers by revenue in descending order	Column / Vertical Bar Chart (sorted desc.)	Clear ranking visual — highest revenue customer immediately visible
Q5	CEO: Product demand across all countries in a single view	Map Chart	Geo-spatial view enables at-a-glance identification of high-demand regions without scrolling

Task 3: Creating Effective Visuals (Power BI)

Data Cleanup Steps

Before analysis, the dataset was cleaned to remove erroneous records:

- Excluded all records where Quantity < 1 (returns recorded as negative quantities)
- Excluded all records where Unit Price < $0 (data entry errors)

These conditional filters ensured the visuals reflected accurate, validated transaction data only.

Visual 1: Revenue Time Series (CEO)

A line chart was built with InvoiceDate on the X-axis (monthly granularity, 2011 only) and Revenue (Quantity * Unit Price) on the Y-axis. The chart revealed a strong upward trend in Q4, with November being the peak revenue month — consistent with holiday retail seasonality.

Visual 2: Top 10 Countries by Revenue & Quantity (CMO)

A side-by-side (clustered) bar chart displays the top 10 countries by revenue, excluding the United Kingdom. Each country shows two bars: total revenue and quantity sold. This dual-metric view lets the CMO compare sales volume efficiency across markets.

Rank	Country	Revenue (approx.)	Key Insight
1	EIRE (Ireland)	£124.72K	Highest revenue outside UK
2	Netherlands	£104.23K	Strong volume + high unit price
3	Germany	£95.39K	Consistent performer
4	France	£73.99K	Growth market
5	Australia	£60.31K	High potential expansion market

Visual 3: Top 10 Customers by Revenue (CMO)

A descending column chart sorted by revenue shows the top 10 customers (identified by CustomerID, with blanks excluded). The highest-revenue customer significantly outpaces the rest, indicating a few key accounts drive disproportionate revenue — a critical insight for retention strategy.

Visual 4: Global Demand Map (CEO)

A filled map chart visualises total units sold by country (excluding the UK). The colour intensity indicates demand level, allowing the CEO to identify expansion opportunities at a glance. Western Europe, Australia, and Japan emerge as high-potential regions.

Task 4: Communicating Insights & Analysis

Presentation Framework

The findings were structured as a 5-minute executive presentation covering:

- Data loading and cleaning methodology (credibility and due diligence)
- Visual 1 insights: Revenue seasonality - peak in Q4, especially November
- Visual 2 insights: Geographic opportunities - EIRE and Netherlands as primary targets
- Visual 3 insights: Customer concentration - top 3 customers contribute 30% of revenue
- Visual 4 insights: Expansion map - Europe-first strategy recommended

Key Recommendations

Area	Recommendation	Priority
Seasonal Strategy	Stock up and launch promotions in Oct–Nov to capitalise on Q4 demand surge	High
Geographic Expansion	Prioritise EIRE, Netherlands, and Germany for expansion investment	High
Customer Retention	Implement a VIP programme for top 10 customers; reduce concentration risk by acquiring mid-tier customers	High
Low-Demand Markets	Review Saudi Arabia and USA market strategy — low volumes suggest pricing or product-fit issues	Medium
Data Quality	Establish ongoing data governance to prevent negative quantity / price errors at source	Medium

Conclusion

The TATA iQ simulation demonstrated a complete end-to-end analytical workflow: from stakeholder question framing to visual selection, data cleaning, Power BI dashboard creation, and executive communication.

The online retail store's revenue is heavily driven by a small number of European markets and a concentrated set of high-value customers. The expansion strategy should prioritise deepening these relationships while diversifying into adjacent geographies with demonstrated demand.

Seasonal trends confirm that Q4 is the critical commercial window - operational readiness and targeted marketing in this period will have the largest impact on annual revenue growth.